Md. Khalequzzaman Sarker Likhon

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Portfolio · GitHub · Hugging Face · LinkedIn · ORCID

RESEARCH INTERESTS

Computer Vision · Video Understanding · Vision-Language Models · Large Language Models · Robustness & Evaluation

EDUCATION

Ahsanullah University of Science and Technology (AUST) Apr 2016 – Jan 2021
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

- CGPA: 2.74/4.00; final-semester GPA: 3.56/4.00, with A grades in Pattern Recognition, Digital Image Processing, and Computer Graphics.
- Thesis: *Design of an Arrhythmia Classification Algorithm Using 2-D Convolutional Neural Networks*; supervisor: Prof. Dr. Mohammad Shafiul Alam.

RESEARCH MANUSCRIPTS

* *denotes equal contribution*

- R. Tanvir*, **M. K. S. Likhon***. Joint Prediction of Ultra-High-Performance Concrete Properties Using a Feature-Tokenizer Transformer with Masked Multi-Task Learning. *Under review, Applied AI Letters (Wiley)*, 2026.
- **M. K. S. Likhon**, M. M. Sarker. Blur Sensitivity and Generator Dependence of Fingerprint Alteration Detectors Trained on SOCOFing. *In preparation*, 2026.
- **M. K. S. Likhon**, M. M. Sarker. Data Audit and Revised Baselines for Structural Damage-Type Classification on PEER Hub ImageNet. *In preparation*, 2026.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Accelx Inc. Jan 2025 – Present
Video understanding, vision-language models, grounded LLM systems *Dhaka, Bangladesh*

- **Violence recognition:** fine-tuned a Kinetics-pretrained R(2+1)D-18 and deployed it for real-time multi-camera monitoring; on RWF-2000 it outperformed optical-flow, pose-fusion, and pose-Transformer models.
- **Model audit:** found that pose-based violence models took their "contact" cues from keypoints missing for both people, and that without identity tracking across frames they could not tell who did what.
- **Zero-shot recognition:** built frozen-OpenCLIP fall and violence recognizers; traced false positives on real footage to viewpoint, lighting, and posture, and turned them into hard-negative prompts (a training-free fix).
- **Grounded alerts:** built a pipeline in which a local LLM writes alerts only from code-computed facts and fine-tuned RoBERTa risk scores, with every citation automatically checked.
- **VLM cascade:** designed an OCR–CLIP–Qwen3-VL cascade that calls the VLM only for hard cases, checks findings against a knowledge base, and flags weak evidence for human review.
- **Weapon detection:** fine-tuned a YOLOv10-L detector for live video streams.

SELECTED RESEARCH PROJECTS

Overcooked AAR: telemetry vs. video for automated after-action review [GitHub] Independent research, 2026

- Built a reproducible pipeline in which Video LLMs write after-action reviews of human–human Overcooked play from the event log or the video, with every timestamped claim checked against the log.
- On 15 episodes, claims from the event log were 67–93% precise vs. 22–43% from video, and video timestamps were no better than random. A 1,646-item perception probe traced the gap: models read the clock and score perfectly, but Qwen2.5-VL identifies held objects only at chance level. Remaining episodes and blind human ratings are ongoing.

DocuVision: key–value extraction from scanned documents [GitHub] [Model] 2026

- Fine-tuned LayoutLMv3-large on FUNSD with a learned question-to-answer linker, raising pair F1 from 0.23 (heuristic matcher) to 0.74 with ground-truth words; released the model on Hugging Face.
- Ablations: even with gold tags, a rule-based linker capped at 0.72 pair F1, and reading-order word grouping cost 25 F1 points on two-column forms.

DocuRoute: retrieval and text-to-SQL routing over SEC filings [GitHub] 2026

- Built an LLM router that sends 10-K questions to hybrid retrieval (BM25 + dense) or sandboxed text-to-SQL and returns cited answers; in evaluation, a general-domain reranker cut faithfulness from 0.93 to 0.77.

DepthCraft: measurable 3D from a single photo [GitHub] 2026

- Lifted Depth Anything V2 metric depth into cleaned point clouds, with a measurement tool that reports each distance with its uncertainty; about 330 ms per image on an RTX A5000.

Sentinel: tracking-based video anomaly detection [GitHub] 2026

- Combined YOLOv8n detection, BoT-SORT tracking, and rule-based event logic for falls, loitering, abandoned objects, crowding, and wrong-way motion; 19.7 FPS end to end on an RTX A5000.

Other projects: semantic segmentation on CamVid (FCN, attention U-Net, DeepLabV3+) [GitHub]; pneumonia detection from chest X-rays [GitHub]; speech-to-speech translation with voice cloning [GitHub]; self-correcting RAG with hallucination auditing [GitHub]. More on my portfolio.

TECHNICAL SKILLS

Methods: action recognition, pose estimation and tracking, object detection, segmentation, VLMs and Video LLMs, document AI, transformer fine-tuning, RAG and LLM grounding, evaluation design, error analysis
Frameworks: PyTorch, Hugging Face Transformers, OpenCLIP, Ultralytics, OpenCV, scikit-learn, Weights & Biases
Engineering: Python, C/C++, SQL, Bash; FastAPI, TorchServe, Docker, Redis, Ollama, Git, Linux
Languages: Bengali (native), English (full professional proficiency)

SERVICE, TEACHING, AND CERTIFICATIONS

Organizing Committee, AUST Codeware programming contest 2019
Home tutor for secondary and higher-secondary students (grades 10–12) 2017 – 2022
Coursera certificates: Neural Networks and Deep Learning; Machine Learning 2021, 2023